

Advanced Topics in Computer Graphics I

## Radiometry and Tone Mapping



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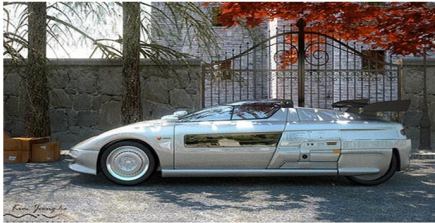
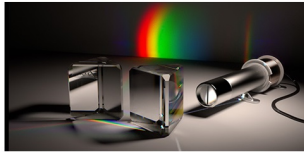
April 15, 2026





**Motivation:** We want to create images of virtual scenes that look photo-realistic.

- ▶ Games / Movies
- ▶ Product Design
- ▶ Architecture

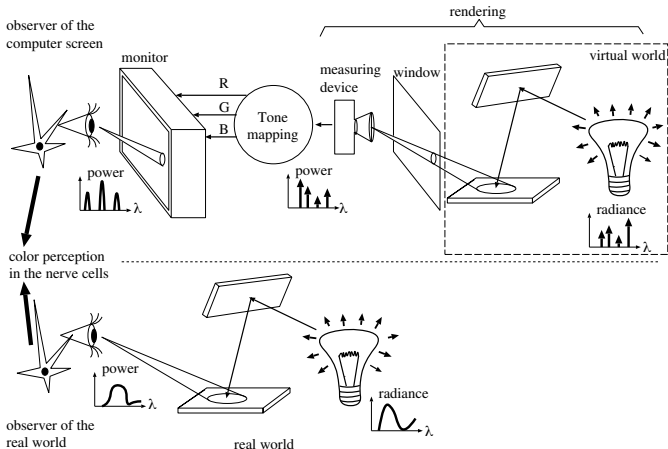




# Rendering Task

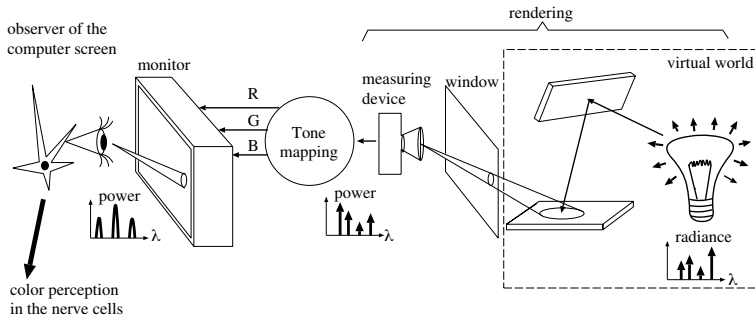


**Task:** Try to simulate the real world!





**Task:** Try to simulate the real world!



There are two parts to consider:

- ▶ Rendering : Light simulation in the virtual world + measuring device
- ▶ Tone Mapping : Measured brightnesses are displayed on a monitor



# Optics

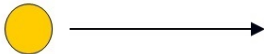




Before we start to consider each part, we need some fundamentals of the physics of light. So we have to answer the question: **What is light?**

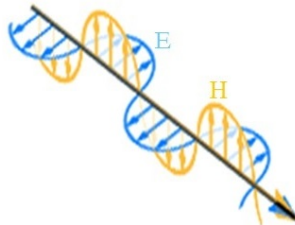
## Light as a particle

- ▶ Light travels along a straight line
- ▶ Light particles = photons



## Light as a wave

- ▶ Electro-magnetic wave
- ▶ Coupled electric (E) and magnetic (H) field

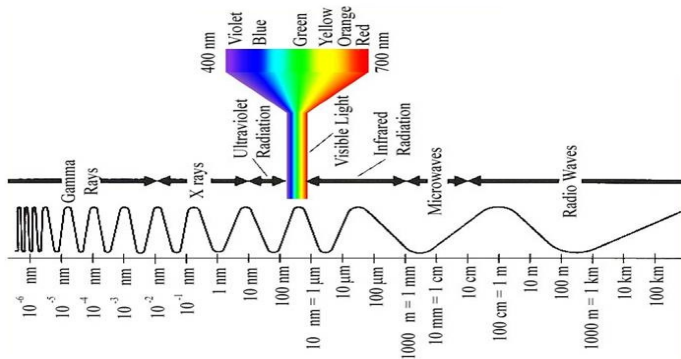




## Wave - Particle Duality:

- ▶ Light has properties of a wave: frequency, phase, orientation
- ▶ Light has properties of a particle: “energy packets” = photons

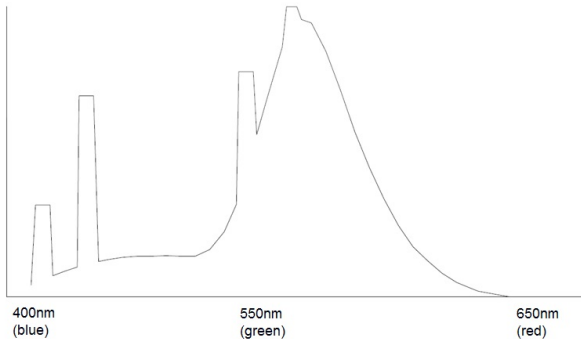
## Visible spectrum:





## Spectral Power Distribution:

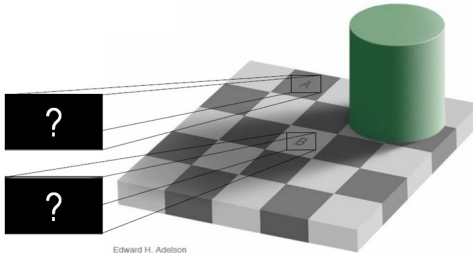
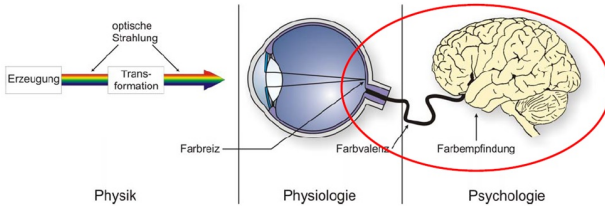
- ▶ Radiation is dependent of wavelength
- ▶ Example: Spectral power distribution of a fluorescent light source (emission spectrum)
- ▶ Total radiated power can be obtained by integration







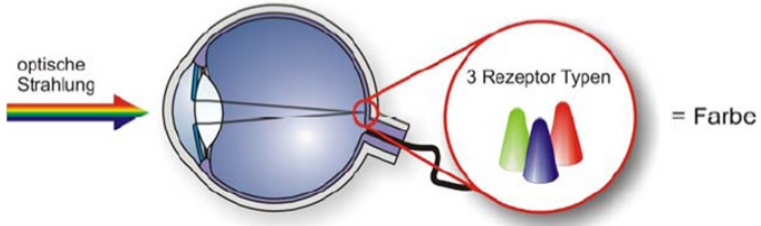
**Observation:** Colors are perceived subjectively → depend on context, person, ...





**Tristimulus Theory:** The perceived color of a spectral distribution can be described completely by three values

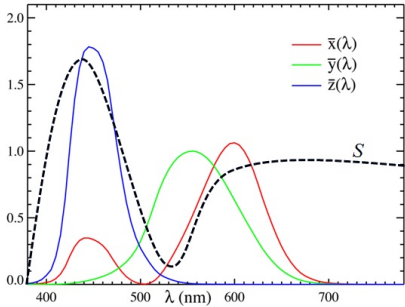
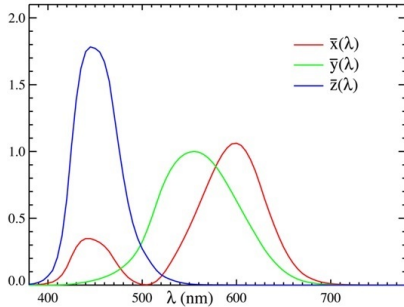
**Reason:** Spectral sensitivity of the three cone cell types (light receptors for color vision)



But what are these colors?



**Cone cell color distributions:** Spectral sensitivity (spectral matching curves) of the receptors according to CIE 1931 standard

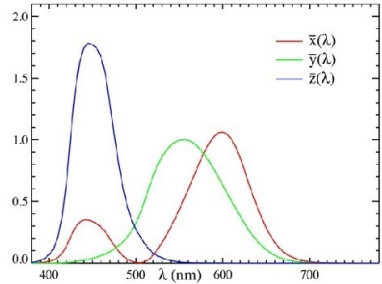
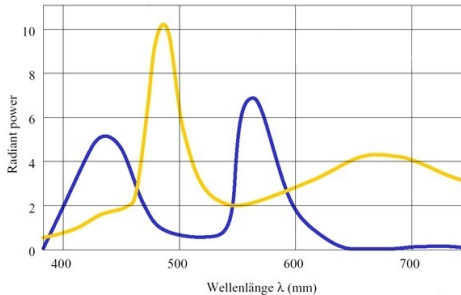


The colors can be computed by projecting the spectrum  $S$  to the sensitivity curves:

$$x = \int_{\lambda} S(\lambda) \bar{x}(\lambda) d\lambda \quad , \quad y = \int_{\lambda} S(\lambda) \bar{y}(\lambda) d\lambda \quad , \quad z = \int_{\lambda} S(\lambda) \bar{z}(\lambda) d\lambda$$



**Consequence:** One perceived color can be generated by different spectral distributions  
→ “metamerism”



Both spectra on the left produce the same  $x, y, z$  colors.



# Radiometry





Now we can define how light is transported through a scene.

- ▶ Radiometry is the science of the measurement of electromagnetic radiation
- ▶ Important radiometric quantities: radiant power, radiant flux, irradiance
- ▶ All these quantities can be represented by spectral distributions!

More precisely, we want to define relations of physical quantities to do the following:

- ▶ Measure the energy (as a function of direction) leaving a light source
- ▶ Measure the energy arriving at a surface from a specific direction
- ▶ Measure the energy leaving surface into a certain direction

**Important assumption:** We neglect the wave nature of the light (and all its effects):

$$\text{Energy} = \text{Photons (Light)}$$



# Solid Angle

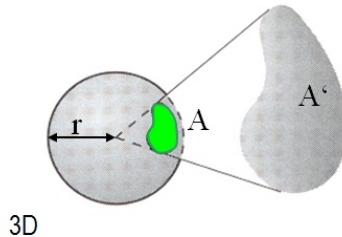
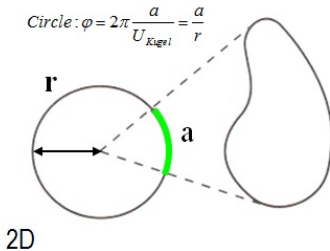


First, we need to define the 2D analogue of an angle.

## Definition (Solid Angle)

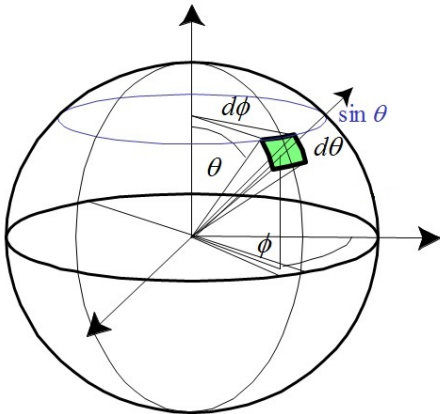
The *solid angle*  $\omega$  is defined as the projected area  $A$  normalized to the unit sphere:

$$\omega = 4\pi \frac{A}{A_{\text{sphere}}} = \frac{A}{r^2} [\text{sr}]$$





In spherical coordinates, we can express the solid angle easily.



The differential surface area (green area) is given as

$$\begin{aligned} dA &= (r d\theta) \cdot (r \sin(\theta) d\phi) \\ &= r^2 \sin(\theta) d\theta d\phi \end{aligned}$$

If we plug this in the definition

$$\omega = \frac{A}{r^2}$$

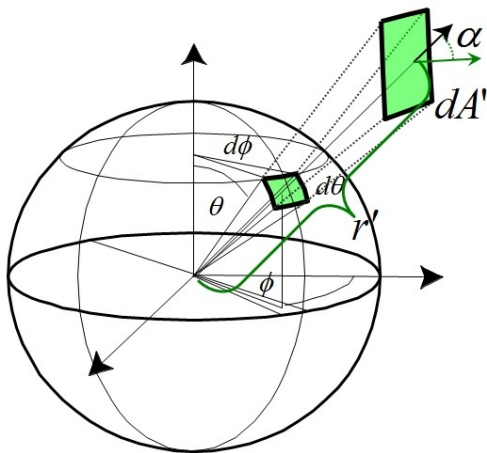
we finally get

$$d\omega = \sin(\theta) d\theta d\phi$$





We can further extend this to surface patches lying far away.



The projected area has the size

$$dA = \cos(\alpha) dA'$$

with a distance of  $r'$ . By definition, we get

$$d\omega = \frac{\cos(\alpha)}{r'^2} dA'$$

For a macroscopic surface element, the induced solid angle can be obtained by integration

$$\omega = \int_{A'} \frac{\cos(\alpha)}{r'^2} dA'$$

### Definition (Radiant Energy)

The *radiant energy* is the basic quantity of the radiometry

$$Q_e \text{ [J]}$$

and can be represented for each wavelength  $\lambda$  individually.

→  $Q_e \propto$  number of photons

### Definition (Radiant Power)

The *radiant power* or *radiant flux* is the radiant energy per time unit

$$\Phi_e = \frac{dQ_e}{dt} \left[ \frac{\text{J}}{\text{s}} = \text{W} \right]$$

and can be represented for each wavelength  $\lambda$  individually.

→  $\Phi_e \propto$  number of photons passing through a surface per time



## Definition (Intensity)

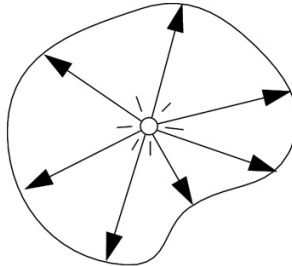
The *intensity* is the radiant power per solid angle

$$I_e = \frac{d\Phi_e}{d\omega} \left[ \frac{\text{W}}{\text{sr}} \right]$$

and can be represented for each wavelength  $\lambda$  individually.

**Example:** Isotropic case

$$I_e = \frac{\Phi_e}{4\pi}$$



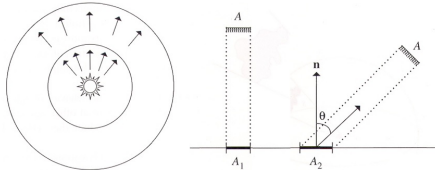


## Definition (Irradiance)

The *irradiance* is the radiant power per area

$$E_e = \frac{d\Phi_e}{dA} \left[ \frac{\text{W}}{\text{m}^2} \right]$$

and can be represented for each wavelength  $\lambda$  individually.



**Example:** Isotropic case

$$E_e = \frac{\Phi_e}{4\pi r^2} \quad , \quad E_e(A_1) = \frac{\Phi_e}{A} \quad , \quad E_e(A_2) = \frac{\Phi_e \cos(\theta)}{A}$$



## Definition (Radiance)

The radiance is the radiant power per area and solid angle

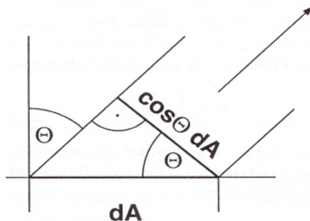
$$L_e = \frac{d^2\Phi_e}{dA d\omega} \left[ \frac{\text{W}}{\text{m}^2 \text{ sr}} \right]$$

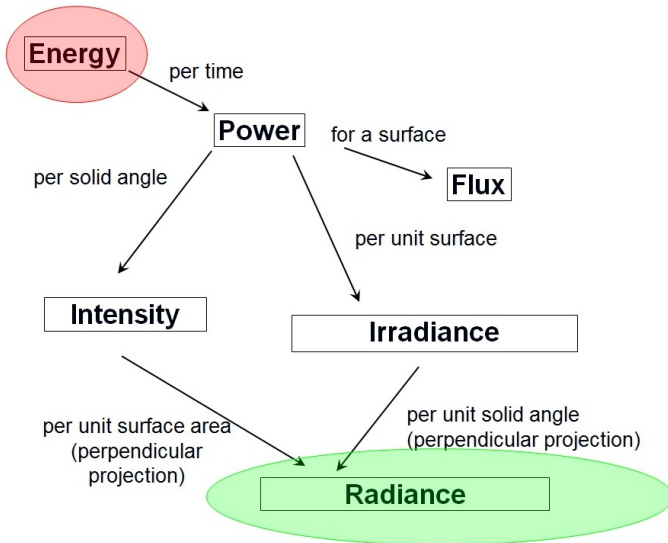
and can be represented for each wavelength  $\lambda$  individually.

The differential area  $dA$  is the area seen from the solid angle  $d\omega$ .

If the surface is not perpendicular, we have an additional term:

$$d^2\Phi_e = L_e \cos(\theta) dA d\omega$$

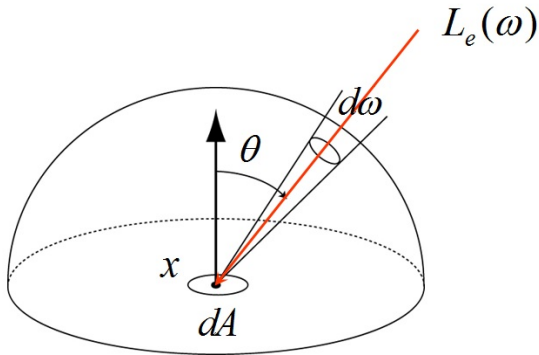






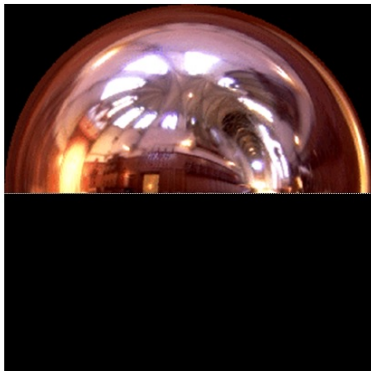
Radiance can be seen as the angular density of irradiance.

$$E_e = \int_{\mathcal{H}} L_e(\omega) \cos(\theta(\omega)) d\omega$$





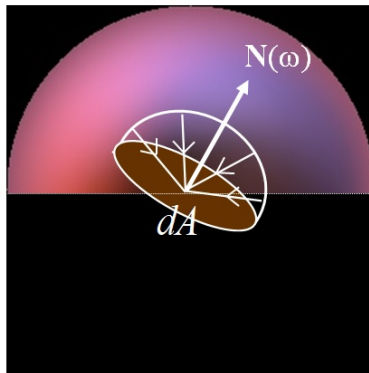
$$L_e(\omega)$$



Radiance



$$E_e(\omega)$$

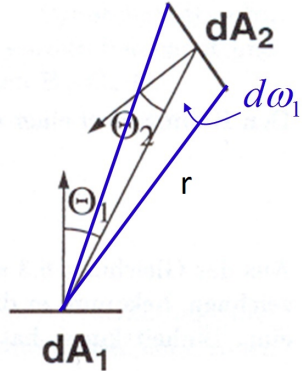


Irradiance



**Consider:** Relation of the differential radiation power, emitted by a differential surface patch  $dA_1$ , and the power, received by a differential surface patch  $dA_2$  having distance  $r$  from  $dA_1$

$$\begin{aligned} d^2\Phi_{e1} &= L_{e1} \cos(\theta_1) dA_1 d\omega_1 \\ &= L_{e1} \cos(\theta_1) dA_1 \frac{\cos(\theta_2)}{r^2} dA_2 \\ &= L_{e1} \frac{\cos(\theta_1) \cos(\theta_2)}{r^2} dA_1 dA_2 \end{aligned}$$



In vacuum, we have energy conservation:

$$d^2\Phi_{e1} = d^2\Phi_{e2}$$

This leads to

$$d^2\Phi_{e1} = L_{e1} \frac{\cos(\theta_1) \cos(\theta_2)}{r^2} dA_1 dA_2$$

$$d^2\Phi_{e2} = L_{e2} \frac{\cos(\theta_1) \cos(\theta_2)}{r^2} dA_1 dA_2$$

so we get the final equation

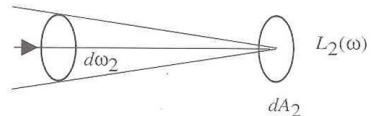
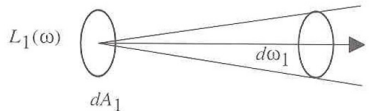
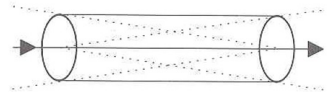
$$L_{e1} = L_{e2}$$

## Theorem (Radiation Transfer)

*The radiance along a ray is constant in vacuum:*

$$L_{e1} = L_{e2}$$

→ Exploited in all rendering algorithms!



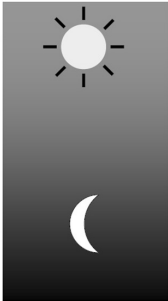


## Dynamic Range





## The range of lighting

|                                                                                  |                       |             |                                              |
|----------------------------------------------------------------------------------|-----------------------|-------------|----------------------------------------------|
|  | Direct sun            | 100'000 Lux | ] Electronic imagers<br>Total lighting range |
|                                                                                  | Sunny day             | 50'000 Lux  |                                              |
|                                                                                  | Cloudy day            | 5'000 Lux   |                                              |
|                                                                                  | Office                | 400 Lux     |                                              |
|                                                                                  | Home lighting         | 10 Lux      |                                              |
|                                                                                  | Street lamps          | 1 Lux       |                                              |
|                                                                                  | Full moon             | 0.1 Lux     |                                              |
|                                                                                  | Quarter moon          | 0.01 Lux    |                                              |
|                                                                                  | Clear moonless night  | 0.001 Lux   |                                              |
|                                                                                  | Cloudy moonless night | 0.0001 Lux  |                                              |

### Problem:

- ▶ Perceivable dynamic range:  
 $\approx 10^7$  values
- ▶ Typical cameras / displays:  
**256 ... 4096** values
  - ▶ 8 bit in consumer cameras / monitors
  - ▶ 10, 12, 14 bit in good DSLRs / HDR monitors

### Questions:

- ▶ How to capture “High Dynamic Range (HDR)” scenes with “Low Dynamic Range (LDR)” cameras?
- ▶ How to map HDR data to “Low Dynamic Range (LDR)” displays?



Figure: Environment maps (image based lighting, IBL)



Figure: Glossy material highlights



**Example:** Photographic camera with standard CCD sensor:

|                                                                                          |                 |
|------------------------------------------------------------------------------------------|-----------------|
| dynamic range of sensor                                                                  | 1 : 1000        |
| exposure variation, handheld / non-static scene, $\frac{1}{60} s \dots \frac{1}{6000} s$ | 1 : 100         |
| varying aperture $f/2.0 \dots f/22.0$                                                    | 1 : 100         |
| exposure bias / varying "sensitivity"                                                    | 1 : 10          |
| <b>total</b> (sequential)                                                                | 1 : 100,000,000 |
| <b>simultaneous</b> dynamic range still                                                  | 1 : 1000        |

Multiple sensors:

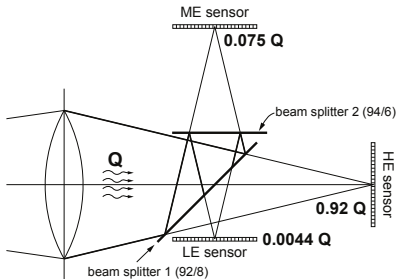


Figure: Using beamsplitters, image scene on multiple sensors at different exposure levels, all at the **same shutter time**! No filters, **high efficiency** (99.96% of light on sensors). Difficulty: **sensor alignment & costs**. Tocci et al., 2011

Modern CMOS sensors can achieve a dynamic range of  $10^6$  to  $10^8$ .

- ▶ logarithm in analog domain
- ▶ exposure bracketing

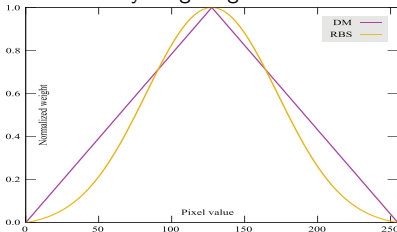
Recently (since 2021) available Qaunta Image Sensors (Photon Counting Cameras at Room Temperature) achieve a dynamic range of  $10^{10}$  (100dB).

Extend dynamic range by combining multiple exposures of the same scene.

## Challenges:



- ▶ cameras are not linear:  $2 \times \text{exposure} \neq 2 \times \text{pixel value}$ 
  - ▶ need to determine camera response function (CRF) in advance
- ▶ deal with under- and overexposure
  - ▶ underexposure: too much noise
  - ▶ overexposure: saturation, blooming
  - ▶ solved by weighting function



+ response curve

linearized images

+ scaling  
+ weighting function



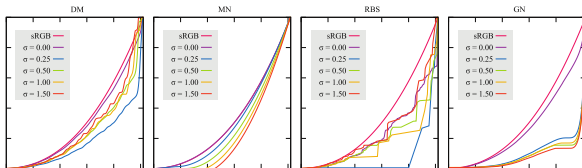
floating point  
HDR image



**Challenge:** Determine camera response function (CRF) / Opto-Electronic Conversion Function (OECF), i.e. the mapping energy to pixel values. Many different approaches:

- ▶ ignore it (assume linear response) [Madden, 1993]
- ▶ optimize for CRF in logarithmic domain under smoothness constraint [Debevec and Malik, 1997]
- ▶ model inverse CRF as polynomial [Mitsunaga and Nayar, 1999]
- ▶ model inverse CRF as independent Gaussian variables [Robertson et al., 2003]
- ▶ assume all CRFs form a subspace, estimate coefficients to basis CRFs [Grossberg and Nayar, 2004]

For a detailed analysis regarding consistency (reproducible?), accuracy and robustness (against noise), please see Akyüz and Gençtav, 2013.

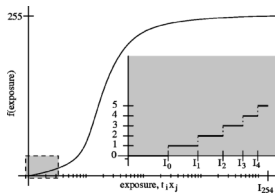




General principle to estimate HDR irradiance map  $x_j$  for all pixels  $j = 1, \dots, M$ :

- assume initial CRF  $f$ , discretized as  $I_m, m = 0, \dots, 255$ , for 8 bits:

$$f(z) = \begin{cases} 0 & \text{if } z \in [0, I_0] \\ m & \text{if } z \in (I_{m-1}, I_m], \\ & m = 1, \dots, 254 \\ 255 & \text{if } z \in (I_{254}, \infty) \end{cases}$$



- input  $N$  LDR images  $y_{ij}$  with different exposure times  $t_i, i = 1, \dots, N$ , where

$$y_{ij} = f(t_i x_j)$$

- Minimize objective for unknown  $I_m$  (i.e. “inverse” of  $f$ ) and  $x_j$ :

$$O(\mathbf{I}, \mathbf{x}) = \sum_{i,j} w_{ij} (I_{y_{ij}} - t_i x_j)^2$$

Additional constraint necessary to prevent arbitrary scale:  $I_{128} \stackrel{!}{=} 1.0 \Rightarrow \hat{I}_m := \frac{I_m}{I_{128}}$



Minimize objective for unknown  $I_m$  (i.e. "inverse" of  $f$ ) and  $x_j$ :

$$O(\mathbf{I}, \mathbf{x}) = \sum_{i,j} w_{ij} (I_{y_{ij}} - t_i x_j)^2$$

Iterate until convergence:

1. calculate irradiance map  $x_j$  using current estimate of inverse CRF  $I$

$$\hat{x}_j = \frac{\sum_i w_{ij} t_i I_{y_{ij}}}{\sum_i w_{ij} t_i^2}$$

2. find better CRF using irradiance map

$$\hat{I}_m = \frac{1}{|E_m|} \sum_{(i,j) \in E_m} t_i x_j,$$
$$E_m = \{(i,j) : y_{ij} = m\}$$

Weights are defined as

$$w_{ij}(y_{ij}) = \exp\left(-4 \frac{(y_{ij} - 127.5)^2}{(127.5)^2}\right), \text{ where } w_{ij}(0) = w_{ij}(255) = 0,$$

and model uncertainty of values close to under- and overexposure.

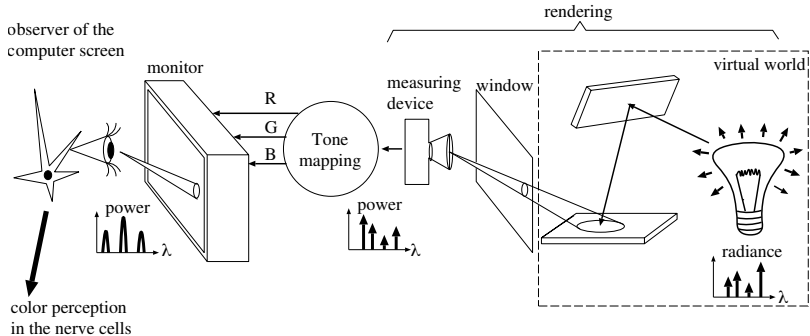


# Tonemapping





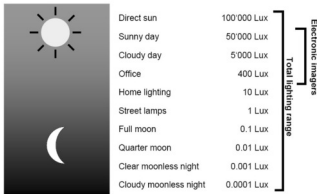
Since we know what light is and how it can be quantified, we come back to the initial problem:



**First Question:** How is Tone Mapping performed?



## The range of lighting



## Problem:

- ▶ Perceivable dynamic range:  $\approx 10^7$  values
- ▶ Typical cameras / displays: 256 ... 16 384 values
  - ▶ 8 bit in consumer cameras / monitors
  - ▶ 10, 12, 14 bit in good DSLRs / HDR monitors

## Questions:

- ▶ How to capture “High Dynamic Range (HDR)” scenes with “Low Dynamic Range (LDR)” cameras?
- ▶ How to map HDR data to “Low Dynamic Range (LDR)” displays?

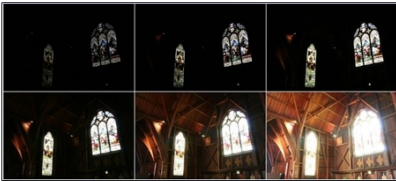
At first glance, Tone Mapping seems rather trivial:

1. Compute radiance hitting the virtual camera ( $\rightarrow$  ranging over the visible spectrum)
2. Compute the  $(x, y, z)$  values using Tristimulus Theory
3. Convert to  $(R, G, B)$  values



## Task: Conversion from HDR to LDR

- ▶ Compress the dynamic range of an input image (real or rendered)
- ▶ Reproduce human perception to closely match that of the real scene
  - ▶ Adaptation of the eye to environment
  - ▶ Contrast
  - ▶ Brightness
  - ▶ Other issues (glare, color perception, resolution)



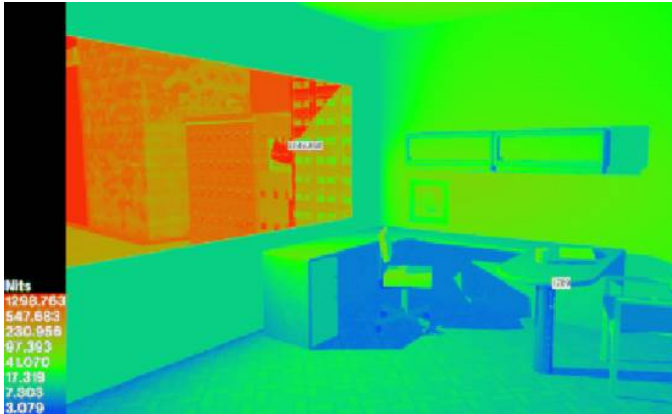
HDR data



LDR display / camera



**Given:** An image consisting of luminance values:



**Note:** There is a high range between the luminance values





**Idea:** Simply scale all values by a factor  $s$  such that they fit into the desired range  $[0, 1]$

$$s = \frac{1}{L_{w \max}}$$

where  $L_{w \max}$  is the maximum luminance of the image.



**Problem:** Light sources are often several orders of magnitude brighter than the rest  
→ rest will be black



**Idea:** Scale to the maximum non-light-source brightness  $L_{w \max nls}$

$$s = \frac{1 - \delta}{L_{w \max nls}}$$

where  $\delta$  is some small constant. Additionally, cap all values above 1.



**Problem:** Brighter regions will be white

**Idea:** Define the scaling factor  $s$  in a way that the visible contrast differences of the image are maintained.

**Goal:** Scale the luminance values such that two values in the real world, that are just noticeably different for an observer, are also just noticeably different on a monitor.<sup>1</sup>

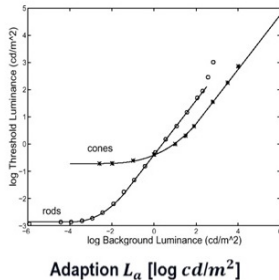
- ▶ Scalar increasing JND would waste the dynamic range of the display
- ▶ Scalar decreasing JND would cause visibly detectable features to disappear

So we need to measure the Just Noticeable Difference (JND):

$$\Delta L(L_a) = 0.0594 (1.219 + L_a^{0.4})^{2.5}$$

where  $L_a$  is the adaption level of the eye (luminance)

Threshold  
 $\Delta L$   
[log cd/m<sup>2</sup>]



<sup>1</sup>International Commission on Illumination. *An Analytic Model for Describing the Influence of Lighting Parameters Upon Visual Performance*. Publication CIE Bd. 1. Bureau Central de la CIE, 1981.

So we need to measure the Just Noticeable Difference (JND):

$$\Delta L(L_a) = 0.0594 (1.219 + L_a^{0.4})^{2.5}$$

where  $L_a$  is the adaption level of the eye (luminance).

- ▶ Luminance  $L_a + \Delta L(L_a)$  can be distinguished from background luminance  $L_a$
- ▶ Luminance  $L_a + \epsilon$  with  $\epsilon < \Delta L(L_a)$  can not be distinguished from background luminance  $L_a$

Now, we want to have a scaling factor  $m$  such that

$$L_d = m \cdot L_w$$

the world luminance  $L_w$  is mapped to the display luminance  $L_d$ .

The key assumption now is that

$$\Delta L(L_d) = m \cdot \Delta L(L_w)$$

where  $\Delta L(L_d)$  is the display and  $\Delta L(L_w)$  the world adaption luminance.

The key assumption now is that

$$\Delta L(L_{da}) = m \cdot \Delta L(L_{wa})$$

where  $\Delta L(L_{da})$  is the display and  $\Delta L(L_{wa})$  the world adaption luminance.

Plugging in the definition of the adaption luminance and solving for  $m$  leads to

$$m = \left( \frac{1.219 + L_{da}^{0.4}}{1.219 + L_{wa}^{0.4}} \right)^{2.5}$$

This scalar converts world luminance to display luminance, so we additionally need to add another term to get a range of  $[0, 1]$ . Furthermore, we assume that the display adaption level can be approximated by  $L_{da} \approx L_{dmax} / 2$ :

$$s = \frac{1}{L_{dmax}} \left( \frac{1.219 + (L_{dmax} / 2)^{0.4}}{1.219 + L_{wa}^{0.4}} \right)^{2.5}$$

The remaining parameters are  $L_{dmax}$ , which is  $250 \text{ cd/m}^2$  for a typical modern LCD display, and  $L_{wa}$ , which has to be estimated from the image.

**Results:** Contrast-based Scaling Factor



**Problem:** Same as for the other uniform scaling approaches



## Problems of simple uniform scaling approaches:

- ▶ Large range of sparsely populated real world luminance levels are mapped to a large range of display values
- ▶ The eye is only sensitive to the fact that there are bright areas and dim areas and not to absolute brightnesses
- ▶ Eye adapts rapidly to the brightness level in a approx  $1^\circ$  visual field around the fixation point

To overcome this, Histogram Adjustment can be performed.<sup>2</sup>

### Step 1: Compute an Adjustment Image

- ▶ Assumes known view point
- ▶ Filtering non-overlapping regions covering  $1^\circ$  fov
- ▶ Box filter (used in the paper) or similar



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<sup>2</sup>G. W. Larson et al. "A visibility matching tone reproduction operator for high dynamic range scenes". In: *Visualization and Computer Graphics, IEEE Transactions on* 3.4 (1997), pp. 291–306.



**Step 2:** Convert world luminance values  $L_w$  to brightness values  $B_w$

- ▶ Tries to model human perception
- ▶ Only crude approximation (i.e. the subjective response)

$$B_w = \log L_w$$

**Step 3:** Compute a histogram of brightnesses

- ▶ Represents sparse distribution of brightness values
- ▶ Number of bins define resolution of clustering, but is not so important if chosen high enough (100 used in the paper)







**Step 4:** Compute a cumulative distribution of the histogram:

- ▶  $f(B_{wi})$  is the number of samples in bin  $i$
- ▶ The accumulated probability is

$$\Pr(B_w) = \frac{\sum_{B_{wi} < B_w} f(B_{wi})}{T}$$

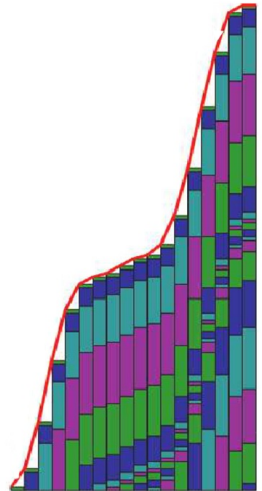
- ▶  $T$  is the total number of samples

$$T = \sum_{B_{wi} < B_{wmax}} f(B_{wi})$$

**Step 5:** Naive Histogram Equalization

- ▶ All brightness values should have equal probability in the final image:

$$B_d = B_{dmin} + (B_{dmax} - B_{dmin}) \cdot \Pr(B_w)$$





## Results:



**Left** : Linear , **Right** : Naive Histogram Equalization (HE)



**Problem:** Too strong emphasis on contrast in highly populated regions of the range

**Idea:** Limit the contrast enhancement (should not exceed that produced by a linear tone mapping operator)

$$\frac{dL_d}{dL_w} \leq \frac{L_d}{L_w}$$

Computing the left-hand side leads to (remember:  $B_w = \log L_w$ )

$$\begin{aligned} \frac{dL_d}{dL_w} &= \frac{d \exp B_d}{dL_w} = \exp B_d \cdot \frac{dB_d}{dL_w} \\ &= \exp B_d \cdot \frac{d}{dL_w} \left( B_{d \min} + (B_{d \max} - B_{d \min}) \cdot \underbrace{\Pr(B_w)}_{=\Pr(\log L_w)} \right) \\ &= \exp B_d \cdot (B_{d \max} - B_{d \min}) \cdot \frac{d\Pr(B_w)}{dL_w} \\ &= \exp B_d \cdot (\log L_{d \max} - \log L_{d \min}) \cdot \frac{d\Pr(B_w)}{dB_w} \cdot \frac{dB_w}{dL_w} \\ &= \exp B_d \cdot (\log L_{d \max} - \log L_{d \min}) \cdot \frac{d\Pr(B_w)}{dB_w} \cdot \frac{1}{L_w} \end{aligned}$$



The only thing left is the derivative of the accumulated probability:

$$\frac{d\Pr(B_w)}{dB_w} \approx \frac{\Pr(B_w + \Delta B_w) - \Pr(B_w)}{\Delta B_w} = \frac{f(B_w)}{T \Delta B_w}$$

So the final result is

$$\frac{dL_d}{dL_w} = \exp B_d \cdot \frac{f(B_w)}{T \Delta B_w} \cdot \frac{\log L_{d \max} - \log L_{d \min}}{L_w} \leq \frac{L_d}{L_w}$$

If we now solve this for  $f(B_w)$ , we get

$$f(B_w) \leq \frac{T \Delta B_w}{\log L_{d \max} - \log L_{d \min}}$$

In other words: As long as no bin count exceeds this ceiling, our histogram will not exaggerate contrast.

**Solution:** Limit the sample count per bin in the histogram



## Results:



**Left** : Naive HE , **Right** : HE with Linear Ceiling



**Further Problem:** Seeing contrasts in dimmed scenes is complicated for the human vision system

**Idea:** Do not only limit the contrast to linear operator, but also to the ratio of the two adaption level thresholds:

$$\frac{dL_d}{dL_w} \leq \frac{\Delta L(L_d)}{\Delta L(L_w)}$$

This guarantees that the display representation does not exhibit contrast that is more noticeable than it would be in the actual scene.

Using the calculations as before, we obtain

$$\frac{dL_d}{dL_w} = \exp B_d \cdot \frac{f(B_w)}{T \Delta B_w} \cdot \frac{\log L_{d \max} - \log L_{d \min}}{L_w} \leq \frac{\Delta L(L_d)}{\Delta L(L_w)}$$

So our new ceiling is given as

$$f(B_w) \leq \frac{\Delta L(L_d)}{\Delta L(L_w)} \cdot \frac{T \Delta B_w}{\log L_{d \max} - \log L_{d \min}} \cdot \frac{L_w}{L_d}$$



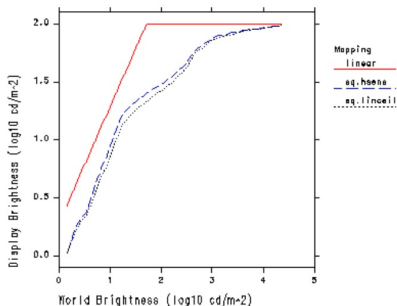
# Histogram Adjustment



## Effect on mapping:

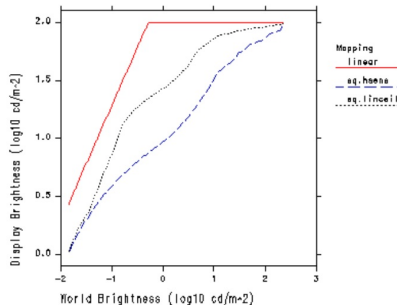
Brightness Mapping Function

Bathroom



Brightness Mapping Function

Dim Bathroom



**Dotted** : Linear ceiling , **Dashed** : JND ceiling



## Results:



**Left** : HE with Linear ceiling , **Right** : HE with JND ceiling





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